UNIVERSITÀ DI TRENTO

Do Algorithms Amplify Financial Inequality?

Evidence from Causal Inference and Machine Learning in FinTech Lending

Abstract

This study examines two related questions about FinTech lending: whether higher interest rates causally increase the probability of default, and whether machine learning credit models treat low-income borrowers unfairly. The data come from LendingClub loans issued between 2007 and 2018.

To address the first question, we use a Fuzzy Regression Discontinuity Design that exploits a natural threshold in the FICO scoring system at 740, where interest rates drop discontinuously. The causal estimate points in the expected direction, as higher rates appear to increase default risk, but the instrument is weak (first-stage $F = 3.31$) and the effect is not statistically significant ($p = 0.751$), so no firm causal conclusion can be drawn.

For the fairness audit, three classifiers, Logistic Regression, Random Forest, and XGBoost, are evaluated on standard fairness metrics stratified by income group. All three models systematically disadvantage low-income borrowers, with Logistic Regression producing the largest gaps. Most importantly, a conditional analysis within FICO score quintiles reveals that the models over-predict default risk for low-income borrowers by 1.16 to 4.06 percentage points even after credit quality is held fixed, a pattern that grows stronger as creditworthiness improves, and that standard regulatory audits would fail to detect.

1. Introduction

The rapid expansion of financial technology has changed how credit is priced, allocated, and denied. Over the past decade, peer-to-peer lending platforms and other FinTech intermediaries have extended credit to millions of borrowers who were previously underserved by traditional banks, promising lower costs, faster decisions, and greater inclusivity. Yet this wave of innovation carries an underexplored tension: the algorithms that power these platforms may replicate, or even amplify, the very inequalities they were designed to dissolve.

This paper sits at the intersection of two distinct but complementary research agendas. The first concerns the causal effect of interest rates on borrower default. When a platform assigns a higher rate to a marginal borrower, does that pricing decision mechanically increase the probability of eventual non-payment, or are the two outcomes merely correlated through shared credit risk? Disentangling cause from correlation is methodologically demanding, because interest rates and default risk are jointly determined by the same underlying borrower characteristics. The second agenda concerns algorithmic fairness: even if a model predicts default accurately on average, it may do so by implicitly weighting income-correlated signals in ways that produce systematically biased predictions against low-income borrowers, independently of their actual credit risk.

To address both questions, this paper uses the LendingClub loan dataset covering the period 2007 to 2018, one of the largest publicly available records of peer-to-peer consumer lending in the United States. The empirical strategy for the causal question relies on a Fuzzy Regression Discontinuity Design (Fuzzy RDD), which instruments for the assigned interest rate using the discrete discontinuity in lending conditions that arises at a FICO score of 740. As documented in a related credit market context by Agarwal et al. (2018), this threshold corresponds to a sharp change in lenders' propensity to extend favourable credit terms, generating quasi-experimental variation in rates near the cutoff. Figure 1 plots average interest rates across the full FICO range and shows two institutional thresholds: an entry cutoff at 660, where borrowers first gain access to the platform, and a prime cutoff at 740, where a visible downward shift in rates motivates its use as an instrument in a two-stage estimation framework. The resulting Local Average Treatment Effect (LATE) is designed to quantify the causal impact of a one-percentage-point increase in interest rate on the probability of default for borrowers near the threshold, though, as discussed in Sections 5 and 6, the strength of the instrument places important qualifications on that estimate.

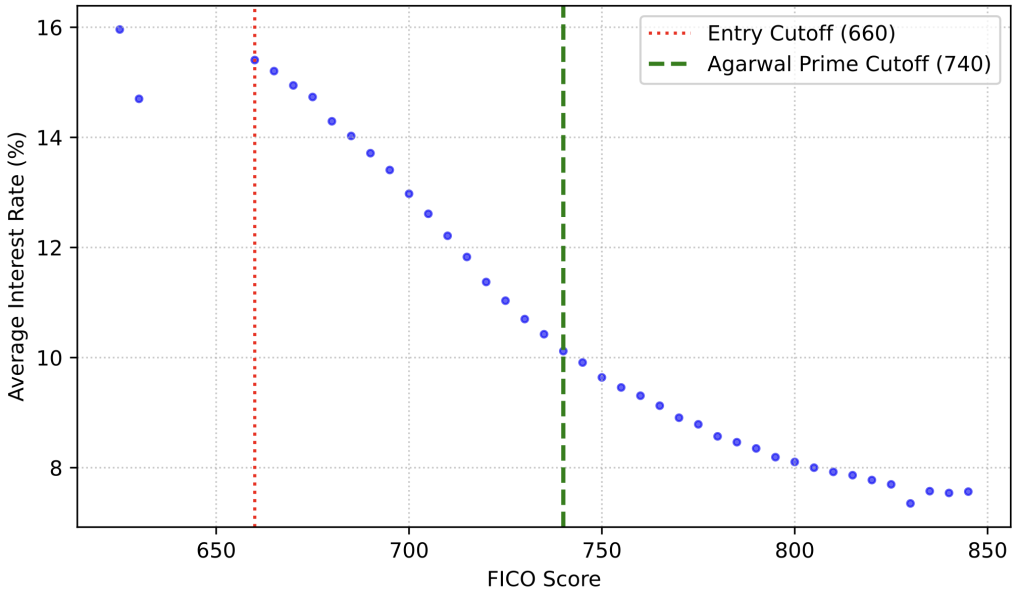

Figure 1: Discontinuity in Interest Rate

For the fairness audit, the paper trains three classifiers of increasing complexity, logistic regression, random forest, and XGBoost, and evaluates each against formal group-fairness criteria drawn from Barocas, Hardt, and Narayanan (2023): Demographic Parity, Equal Opportunity, and Predictive Parity. Building on Fuster et al. (2022), who demonstrate that machine learning models in credit markets can generate predictable and systematic disparities across borrower groups, the audit stratifies all fairness metrics by income group and uses SHAP decompositions (Lundberg and Lee, 2017) to examine how each model weights income-correlated features. A conditional fairness analysis then tests whether borrowers with identical FICO scores receive materially different predicted default probabilities depending solely on their income bracket, which is the empirical signature of what Bartlett et al. (2022) identify as algorithmic proxy discrimination in FinTech lending markets.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature on causal mechanisms in credit markets, algorithmic fairness, and proxy discrimination. Section 3 formalizes the research questions and testable hypotheses. Section 4 describes the data and the empirical methodology. Section 5 presents results across both the causal and the predictive dimensions. Section 6 discusses limitations and threats to validity. Section 7 synthesizes the findings, draws policy implications, and outlines directions for future research.

2. Background and Related Work

This study draws on three areas of research: the causal analysis of credit market mechanisms, the fairness implications of machine learning in financial decision-making, and the specific challenge of proxy discrimination in FinTech lending.

2.1 Causal Mechanisms in Credit Markets

Understanding the causal relationship between credit conditions and borrower outcomes has long been complicated by a fundamental identification problem: interest rates and default risk are simultaneously determined by the same underlying borrower characteristics, making it difficult to isolate the independent effect of pricing on repayment behaviour. A strand of the empirical finance literature has addressed this challenge by exploiting discontinuities in lenders' decision rules as sources of quasi-experimental variation.

Agarwal et al. (2018), in a related credit market context, show how discrete thresholds embedded in lenders' internal scoring systems generate sharp changes in the credit conditions extended to borrowers, changes that are effectively exogenous to individual creditworthiness near the cutoff. This local randomisation supports causal inference without experimental manipulation. Whether this mechanism applies with equal force to the LendingClub context examined here is an empirical question: the FICO 740 threshold produces a visible discontinuity in average interest rates (Figure 1), but the strength of the first stage remains to be established.

The literature has further documented that pricing decisions themselves can alter repayment outcomes through a mechanical debt-burden channel: higher interest rates increase the total repayment obligation, raising the probability that a borrower's income becomes insufficient to service the debt. This channel predicts a positive causal effect of interest rates on default, which is the hypothesis the RDD strategy in this paper is designed to test.

2.2 Algorithmic Fairness and Machine Learning

Parallel to the causal literature, a growing body of scholarship has examined the distributional implications of replacing traditional statistical credit models with high-dimensional machine learning algorithms. Fuster et al. (2022) demonstrate that this transition is not distributionally neutral: analysing mortgage applications across the United States, they show that machine learning models generate systematically larger disparities in predicted risk across racial and ethnic groups than their parametric predecessors, even when trained on identical features. The mechanism is important: complex models amplify the predictive signal embedded in income and geography-correlated variables, effectively translating socioeconomic inequality in the training data into inequality in credit pricing, while aggregate accuracy metrics remain largely unchanged.

This observation motivates the adoption of formal group-fairness criteria as evaluation standards complementary to predictive accuracy. The framework developed by Barocas, Hardt, and Narayanan (2023) distinguishes among Demographic Parity, Equal Opportunity, and Predictive Parity, three non-equivalent definitions that cannot be simultaneously satisfied when base rates differ across groups, implying that any fairness audit must make explicit which criterion is prioritised, and this paper evaluates all three. The analysis stratifies by income rather than race due to data availability constraints, and formally defines each measure in Section 4.3.

2.3 Proxy Discrimination in FinTech

FinTech platforms are legally prohibited from incorporating protected characteristics such as race, ethnicity, or gender directly into their credit models. However, as Bartlett et al. (2022) document using Home Mortgage Disclosure Act data, this legal constraint does not eliminate disparate treatment. Latino and Black borrowers pay materially higher interest rates than observationally equivalent white borrowers even on algorithmic platforms, suggesting that models learn to approximate group membership through correlated features such as ZIP code, employment sector, or loan purpose. These disparities persist after controlling for FICO score and other financial variables. While this paper cannot replicate that analysis due to the absence of race data in the LendingClub dataset, income serves as a closely related stratification variable: it is correlated with race, legally permissible as a model input, and directly observable in the data.

Chen (2024) describes this broader phenomenon as Algorithmic Proxy Discrimination: neutral input variables can become discriminatory when they act as statistical proxies for protected group membership. Chen argues that the standard regulatory response, which focuses on auditing whether protected attributes are explicitly included in a model, is not sufficient in settings where complex models can effectively reconstruct group membership from seemingly neutral features. The appropriate response is therefore a shift from checking model inputs to evaluating model outputs, which is exactly what the conditional fairness analysis in Section 5.4 of this paper does.

3. Research Question and Hypotheses

This study addresses two gaps in the existing literature. First, it is still unclear whether higher interest rates causally increase the probability of default in peer-to-peer lending. Second, it is not yet established whether algorithmic models treat borrowers differently based on income, even when their credit quality is the same.

3.1 Causal Hypotheses (RQ1)

Research Question 1: Does a lower interest rate at the FICO 740 threshold reduce the probability of default?

This paper uses the discontinuity at FICO 740 to estimate the causal effect of interest rates on default, within a window of ±25 points around the cutoff. Three hypotheses guide this analysis:

- **H1a (First Stage):** Borrowers just above the 740 threshold receive meaningfully lower interest rates than borrowers just below it. If this difference is not strong enough, the causal estimate will not be reliable.
- **H1b (Causal Effect):** A higher interest rate causally increases the probability of default for borrowers near FICO 740. This is tested at three different bandwidths and checked against placebo cutoffs.
- **H1c (Heterogeneity):** The effect in H1b is expected to be stronger for lower-income borrowers, who are more financially vulnerable. Due to smaller subgroup sizes, the focus is on the direction of the effect rather than its statistical significance.

3.2 Algorithmic Fairness Hypotheses (RQ2)

Research Question 2: Do machine learning models produce systematically biased predictions against low-income borrowers, even when their credit quality is held fixed?

Three classifiers are trained and evaluated: logistic regression, random forest, and XGBoost. The fairness audit is structured around three progressively demanding questions, each corresponding to a hypothesis:

- **H2a (Global Fairness):** XGBoost is expected to show larger fairness gaps between low and high-income borrowers than logistic regression, because more complex models tend to amplify income-correlated signals in the training data.
- **H2b (Feature Attribution):** Within the XGBoost model, income is expected to carry disproportionately higher predictive weight for low-income borrowers than for high-income borrowers, suggesting that the model penalises the most disadvantaged group most heavily.
- **H2c (Excess Bias):** Even after holding credit quality fixed within FICO quintiles, XGBoost is expected to assign higher predicted default probabilities to low-income borrowers than to high-income borrowers with equivalent credit scores. The difference between the predicted gap and the actual gap in default rates is defined as excess bias, and constitutes the main empirical test of algorithmic proxy discrimination in this paper.

H2c is the most stringent test: conditioning on FICO quintile removes the primary legitimate predictor of default risk, leaving income-based bias as the only remaining explanation for any residual prediction gap.

4. Analytical Approach

4.1 Data Provenance and Preprocessing

This study uses the LendingClub loan dataset covering the period 2007 to 2018, one of the largest publicly available records of peer-to-peer consumer lending in the United States. Four variables are retained: the lower bound of the borrower's FICO score range, the assigned interest rate, the final loan status, and annual income.

The sample is restricted to loans with a resolved outcome, keeping only observations classified as Fully Paid, Charged Off, or Default, and dropping all loans still active at the time of data extraction. This follows Di Maggio and Yao (2021), who argue that including unresolved loans makes it impossible to measure default accurately. A binary outcome variable is constructed: default equals one if the loan status is Charged Off or Default, and zero if Fully Paid. Observations with missing values are dropped.

Annual income is log-transformed to reduce its right skew, which is standard practice in the credit literature. Borrowers are then split into two groups at the sample median: Low Income for those at or below the median, and High Income for those above, producing two subgroups of roughly equal size.

Table 1 reports descriptive statistics for borrowers within a ±25-point window around the FICO 740 cutoff. The interest rate gap is visible and meaningful: borrowers just below the threshold face a mean rate of 11.20%, compared to 9.68% for those just above, a difference of about 1.5 percentage points. Default rates follow the same pattern, at 14% below and 11% above the cutoff. Mean annual income appears similar across the two groups ($82,052 below versus $83,257 above), though a formal balance test in Section 5.1 reveals a statistically significant income difference at the threshold.

Variable	Statistic	Below Threshold (FICO < 740)	Above Threshold (FICO ≥ 740)
Interest Rate (%)	Mean	11.20	9.68
	Std	4.21	3.81
Annual Income ($)	Mean	82,052	83,257
	Std	78,769	71,814
Default (0/1)	Mean	0.14	0.11
	Std	0.35	0.31

Table 1: Summary Statistics at FICO 740 Cutoff

4.2 Causal Identification Strategy: Fuzzy Regression Discontinuity Design

The main challenge in estimating the effect of interest rates on default is that the two variables are determined by the same underlying borrower characteristics, making it impossible to separate cause from correlation using standard regression. This paper addresses this problem with a Fuzzy Regression Discontinuity Design (Fuzzy RDD), which exploits the discontinuity in lending conditions at a FICO score of 740.

As documented in a related credit market context by Agarwal et al. (2018), borrowers crossing this threshold experience a discrete downward shift in their assigned interest rate. Since borrowers cannot precisely manipulate their FICO score to land on a preferred side of the cutoff, assignment near the threshold is approximately random, which supports a causal interpretation of the estimated effect. The analysis is restricted to borrowers within a ±25-point window around the cutoff [715, 765], where local comparability between groups is most credible. Robustness is assessed across three bandwidth specifications: ±15, ±25, and ±35 points.

The Fuzzy RDD proceeds in two stages.

The First Stage estimates the effect of crossing the FICO 740 threshold on the assigned interest rate. The threshold indicator equals one for borrowers at or above 740 and zero otherwise, and the running variable is centered at the cutoff:

$$int\_rate_i = \alpha_0 + \alpha_1 Z_i + \alpha_2 \tilde{r}_i + \varepsilon_i$$

The coefficient on the threshold indicator captures the jump in interest rates at the cutoff. A strong first stage, conventionally assessed by an F-statistic exceeding 10 on the threshold indicator, is a necessary precondition for reliable causal inference.

The Second Stage estimates the causal effect of the interest rate on the probability of default, using the threshold indicator as an instrument:

$$default_i = \beta_0 + \beta_1 \widehat{int\_rate}_i + \beta_2 \tilde{r}_i + u_i$$

The coefficient on the instrumented interest rate is the Local Average Treatment Effect (LATE): the causal impact of a one-percentage-point increase in interest rate on the probability of default for borrowers near FICO 740. Both stages are estimated jointly via IV2SLS using the linearmodels library, which produces correct standard errors.

Two additional validity checks are reported. A McCrary density test examines whether the distribution of FICO scores shows a discontinuous jump at 740, which would signal manipulation. Placebo tests repeat the first-stage estimation at six alternative cutoffs, 660, 700, 720, 760, 780, and 800, where no institutional discontinuity is expected. A placebo passes if the estimated jump is not statistically significant.

4.3 Predictive Modeling and Fairness Audit Metrics

The fairness audit trains three classifiers on the full analytical sample using three features: FICO score, interest rate, and log-transformed annual income. The dataset is split into training (80%) and test (20%) sets with a fixed random seed to ensure reproducibility. All fairness evaluations are performed on the held-out test set.

Logistic Regression serves as the parametric baseline, estimated with L2 regularisation. Random Forest is an ensemble of 200 trees with a maximum depth of 6. XGBoost is a gradient-boosted ensemble of 200 estimators with maximum depth 4 and learning rate 0.05, with class imbalance corrected via the scale_pos_weight parameter. All three models use balanced class weights. Predictive performance is evaluated via the area under the ROC curve (AUC), computed separately for Low and High Income subgroups.

Three fairness metrics are evaluated at a classification threshold of 0.5, each expressed as the absolute gap between income groups.

Demographic Parity requires equal positive prediction rates across groups, regardless of actual outcomes:

$$\Delta_{DP} = | P(\hat{y} = 1 | Low) - P(\hat{y} = 1 | High) |$$

Equal Opportunity requires equal true positive rates across groups:

$$\Delta_{EO} = | P(\hat{y} = 1 | y = 1, Low) - P(\hat{y} = 1 | y = 1, High) |$$

Predictive Parity requires that among borrowers flagged as high risk, the fraction who actually defaulted does not differ by income group:

$$\Delta_{PP} = | Precision_{Low} - Precision_{High} |$$

SHAP decompositions (Lundberg and Lee, 2017) are computed for the XGBoost model using the TreeExplainer algorithm, applied to a sample of 50,000 test observations. Two quantities are reported: the global mean absolute SHAP value for each feature, and the mean absolute SHAP value for income computed separately for each income group.

Finally, conditional fairness is assessed by partitioning the test set into FICO quintiles and comparing, within each quintile, the average predicted default probability between Low and High Income borrowers. The excess bias is defined as:

$$ExcessBias_q = (\bar{p}_{Low,q} - \bar{p}_{High,q}) - (\bar{y}_{Low,q} - \bar{y}_{High,q})$$

A positive excess bias means the model over-predicts default risk for low-income borrowers beyond what their actual repayment behaviour justifies, which is the empirical signature of algorithmic proxy discrimination in the sense of Chen (2024).

5. Results

5.1 Causal Inference: The Effect of Interest Rates on Default

The Fuzzy RDD estimation uses 273,631 borrowers within the ±25 point bandwidth around FICO 740. The analysis proceeds in three steps: validity checks, first-stage evidence of a rate discontinuity, and the second-stage causal estimate.

Validity checks. Two preconditions must hold for a credible RDD. First, borrowers should not be able to precisely manipulate their FICO score to land just above the threshold. Figure 2 shows the frequency distribution of borrowers within the bandwidth: the histogram declines smoothly and continuously on both sides of 740, with no spike or bunching at the cutoff, which is consistent with the absence of strategic manipulation.

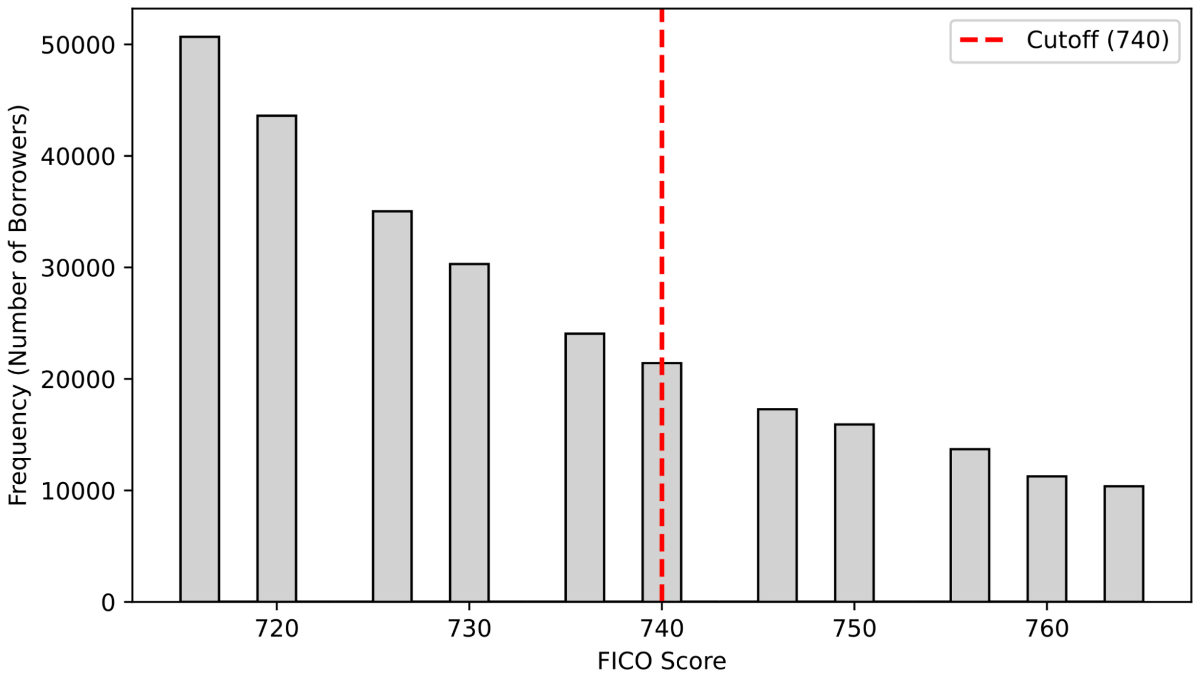

Figure 2: Density Test (McCrary) at Cutoff 740

Second, borrowers just above and just below the threshold should be comparable on pre-determined characteristics. A t-test on log-transformed income rejects this assumption ($t = -2.452$, $p = 0.014$), indicating a statistically significant income difference across the cutoff. This covariate imbalance constitutes a threat to local comparability and is discussed in Section 6.

First stage. Figure 3 shows a visible downward shift in average interest rates at FICO 740, consistent with the institutional mechanism documented by Agarwal et al. (2018). Borrowers just below the cutoff face rates of approximately 10.4 - 10.5%, while those just above are assigned approximately 10.1%. The formal OLS regression estimates the discontinuous jump at the cutoff at −0.059 percentage points (HC3-robust SE = 0.032, p = 0.069), after controlling for the linear FICO trend within the bandwidth. This coefficient captures only the discontinuous shift at the exact threshold and is therefore smaller than the raw mean difference of 1.5 percentage points reported in Table 1, which reflects variation across the full bandwidth. Critically, the first-stage F-statistic is 3.31, well below the conventional threshold of 10. The instrument is weak, and all second-stage estimates must be interpreted with caution.

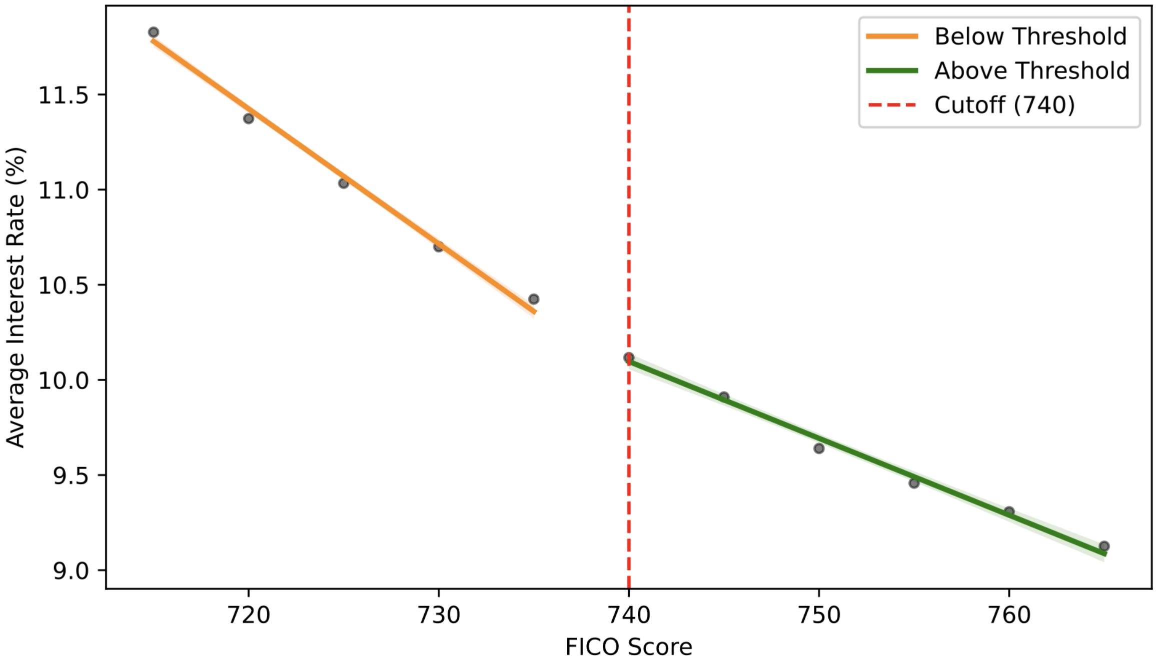

Figure 3: First Stage: Discontinuity in Interest Rate at FICO 740

Second stage. Figure 4 shows a corresponding downward shift in default probability at FICO 740, mirroring the pattern in interest rates. The IV2SLS estimate of the Local Average Treatment Effect is 0.014 (SE = 0.044, p = 0.751): a one-percentage-point exogenous increase in interest rate is associated with a 1.4 percentage-point increase in default probability for borrowers near the cutoff. The sign is consistent with the debt-burden mechanism described in Section 2.1, but the effect is not statistically distinguishable from zero a direct consequence of the weak instrument.

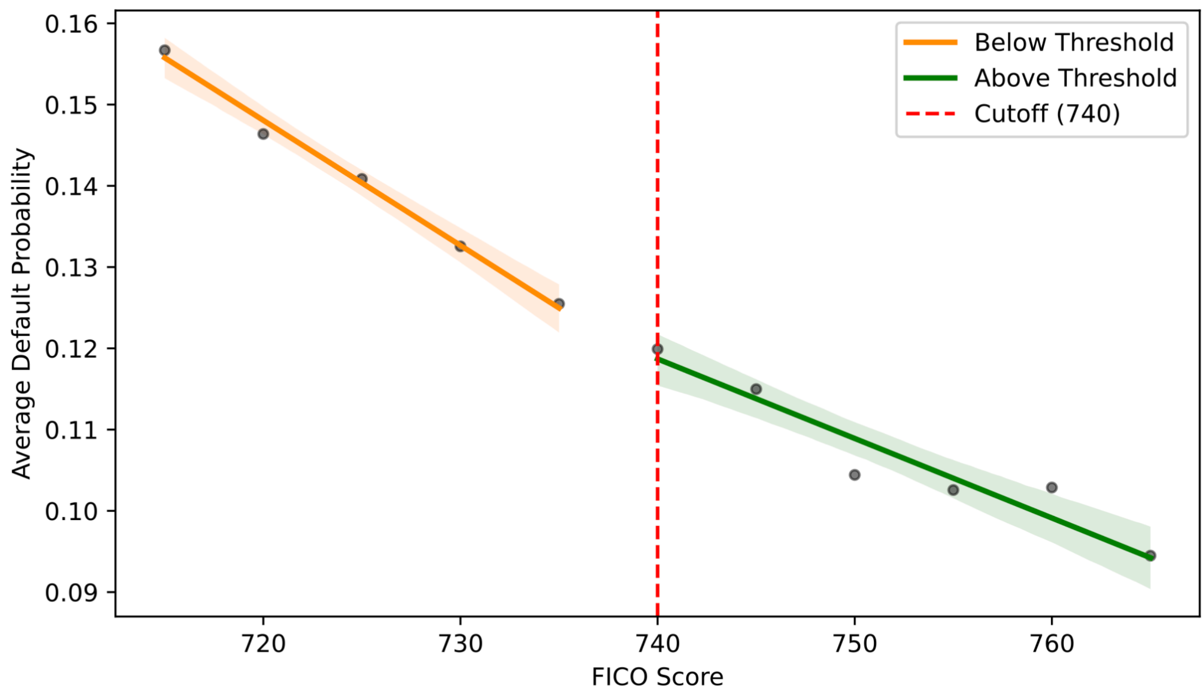

Figure 4: Reduced Form Default Probability at Cutoff 740

Robustness and heterogeneity. Bandwidth sensitivity reveals instability: the LATE is −0.031 at ±15 points (p = 0.715), +0.014 at the baseline ±25 points (p = 0.751), and +0.022 at ±35 points (p = 0.036). The sign reversal between the two narrower specifications indicates sensitivity to the choice of local window, and the only significant estimate at the widest bandwidth is also the one most likely to violate local comparability. Placebo tests at FICO 700, 780, and 800 correctly yield no significant rate discontinuity; cutoffs at 720 and 760 show marginal significance attributable to LendingClub's rate-setting grid; and the entry cutoff at 660 produces a large jump of +7.73 percentage points, confirming that the rate schedule embeds multiple structural discontinuities. The income-stratified analysis finds a LATE of +0.066 for low-income borrowers (p = 0.151) and −0.285 for high-income borrowers (p = 0.723); neither is significant, and the implausibly large high-income estimate reflects the instrument's particular weakness in that subsample. Hypothesis H1c cannot be assessed on these results. Taken together, the causal analysis yields directionally plausible but statistically inconclusive evidence; the specific threats to validity are examined in Section 6.

5.2 Predictive Performance and Global Fairness

The fairness audit evaluates three classifiers on a held-out test set of 269,070 borrowers 139,856 Low Income and 129,214 High Income. In terms of raw discriminative ability, XGBoost achieves the highest overall AUC at 0.690, marginally outperforming Random Forest (0.686) and Logistic Regression (0.685) a difference of half a percentage point, suggesting that model complexity yields limited aggregate predictive gain on this three-feature dataset. Across all three classifiers, the ROC curve for High Income borrowers lies consistently above that for Low Income borrowers, with AUC gaps of 0.024 - 0.025 (Table 2).

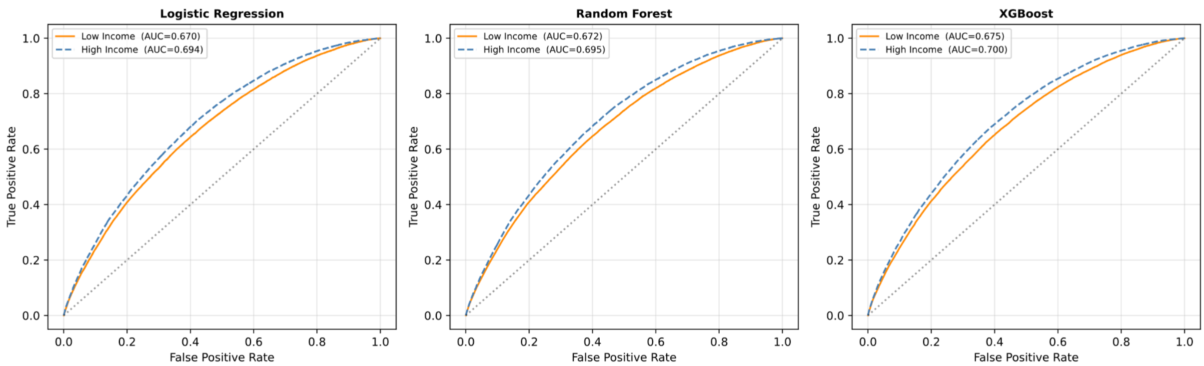

Figure 5: ROC Curves by Income Group

Model	AUC Overall	AUC Low	AUC High	DP Gap	EO Gap	PP Gap
Logistic Regression	0.685	0.670	0.694	0.171	0.146	0.013
Random Forest	0.686	0.672	0.695	0.159	0.114	0.019
XGBoost	0.690	0.675	0.700	0.165	0.124	0.016

Table 2: Summary of Fairness Metrics Across Models

The Demographic Parity and Equal Opportunity gaps are large and consistent across all three models. For XGBoost, 53.9% of Low Income borrowers receive a positive prediction compared to 37.4% of High Income borrowers a gap of 16.5 percentage points. The False Positive Rate gap reaches 0.160, confirming that the over-prediction of risk for low-income borrowers extends to the broader low-income population, not only to actual defaulters. By contrast, Predictive Parity gaps are small across all models (0.013–0.019), indicating that the precision of positive predictions does not differ substantially by income group.

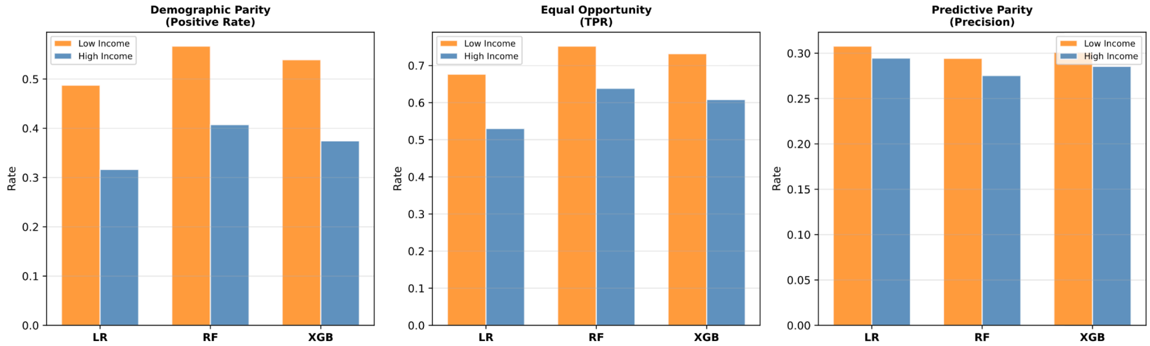

Figure 6: Fairness Metrics by Model and Income Group

	Logistic Regression	Random Forest	XGBoost
Low Income (N = 139,856)			
Pos. Rate (DP)	0.4875	0.5668	0.5392
TPR (EO)	0.6766	0.7523	0.7319
Precision (PP)	0.3078	0.2944	0.3010
FPR	0.4336	0.5139	0.4843
AUC	0.6701	0.6717	0.6751
High Income (N = 129,214)			
Pos. Rate (DP)	0.3164	0.4074	0.3743
TPR (EO)	0.5304	0.6386	0.6081
Precision (PP)	0.2946	0.2754	0.2855
FPR	0.2707	0.3582	0.3245
AUC	0.6941	0.6954	0.6996
Gaps |Low − High|			
Pos. Rate (DP)	0.1711	0.1593	0.1649
TPR (EO)	0.1462	0.1137	0.1238
Precision (PP)	0.0132	0.0190	0.0156
FPR	0.1629	0.1557	0.1598
AUC	0.0239	0.0238	0.0245

Table 3: Detailed Fairness Metrics and Gaps by Income Group

These findings partially contradict Hypothesis H2a, which predicted that XGBoost would exhibit larger fairness gaps than Logistic Regression. The data show the opposite: Logistic Regression produces the largest DP gap (0.171) and EO gap (0.146), while Random Forest produces the smallest on both dimensions, with XGBoost at an intermediate position. This suggests that the fairness problem is not driven by model complexity, but is a structural feature of the income–default relationship in the training data a pattern that all three classifiers learn and reproduce equally. The key finding of H2a nevertheless holds: all three models exhibit large and economically meaningful income-based disparities, and the aggregate AUC gain from Logistic Regression to XGBoost does not translate into any reduction in income-based bias.

5.3 Explainability and Feature Importance (SHAP)

To understand how the XGBoost model arrives at its predictions, SHAP decompositions (Lundberg and Lee, 2017) are computed on a sample of 50,000 test observations. Figure 7 shows the mean absolute SHAP value for each of the three input features.

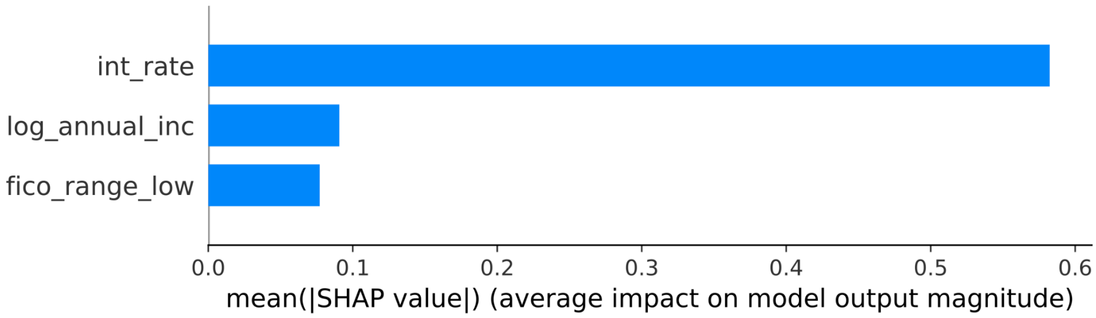

Figure 7: Mean |SHAP|: Global Feature Importance

Interest rate is by far the dominant predictor, with a mean |SHAP| of approximately 0.58 roughly six times larger than log_annual_inc (≈0.10) and fico_range_low (≈0.08). This is consistent with the structure of the dataset: interest rate is the variable that most directly and continuously tracks borrower risk.

Figure 8 disaggregates the SHAP contributions at the individual level, showing not just magnitude but direction.

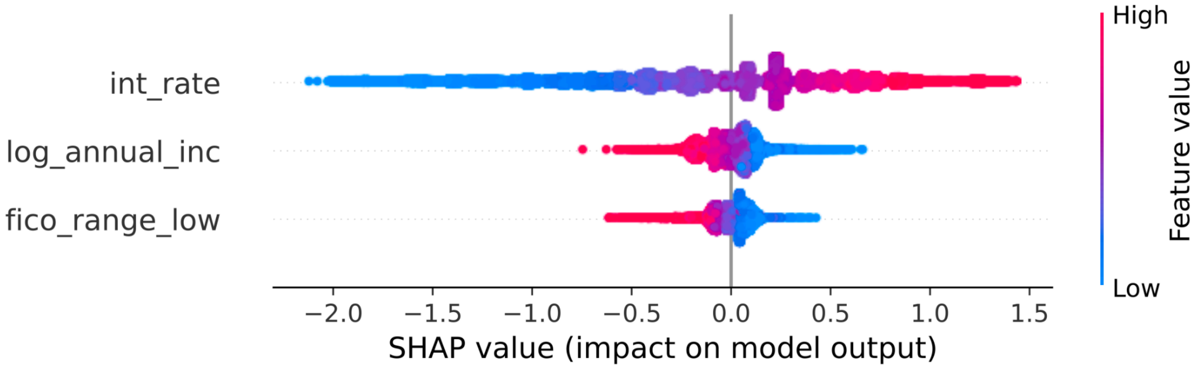

Figure 8: SHAP Beeswarm

For int_rate, the pattern is monotone: high interest rates push predicted default probability upward, low rates push it downward. For log_annual_inc, the direction is inverted high income values reduce predicted default probability, while low income values increase it. This asymmetry is the empirical signature that Chen (2024) associates with algorithmic proxy discrimination: the model systematically uses income to inflate predicted default probabilities for low-income borrowers, producing an effect indistinguishable from explicit group-based pricing.

Figure 9 and Table 4 report mean |SHAP| values separately for Low and High Income borrowers.

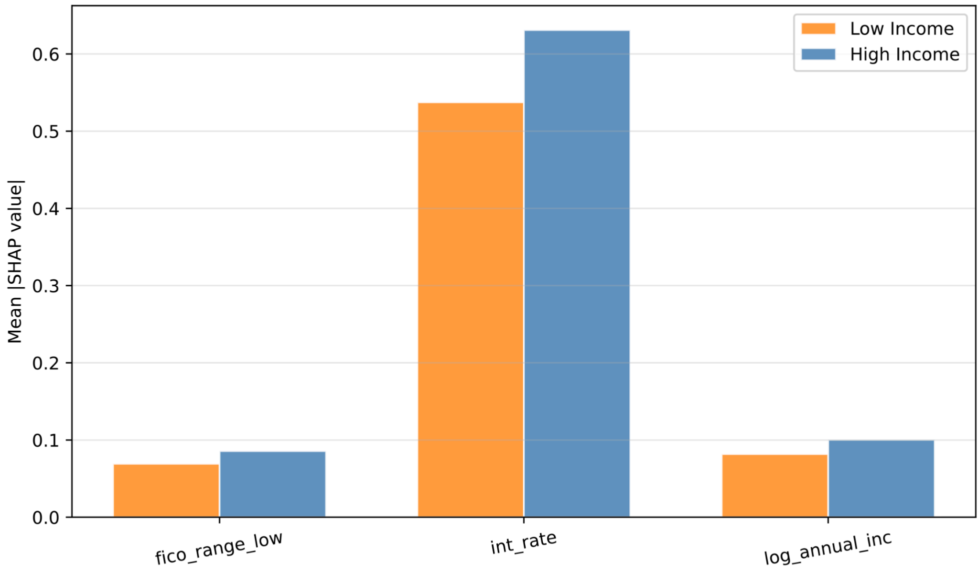

Figure 9: Mean |SHAP| by Feature and Income Group

Feature	Low Income	High Income
fico_range_low	0.069	0.086
int_rate	0.537	0.631
log_annual_inc	0.082	0.100

Table 4: Mean SHAP Values by Income Group

Across all three features, High Income borrowers show higher mean |SHAP| values which does not confirm Hypothesis H2b in its literal form, since income does not carry disproportionately greater weight for low-income borrowers. The relevant finding, however, is directional rather than about magnitude: income penalises low-income borrowers by pushing their predictions upward, while it protects high-income borrowers by pulling their predictions downward. The same feature operates in structurally opposite ways for the two groups.

5.4 Deep Dive: Conditional Fairness and Excess Bias

The most demanding test of algorithmic bias asks whether income-based prediction gaps persist once credit quality is held fixed. Figure 10 compares predicted and actual default rates between Low and High Income borrowers within each of the five FICO quintiles.

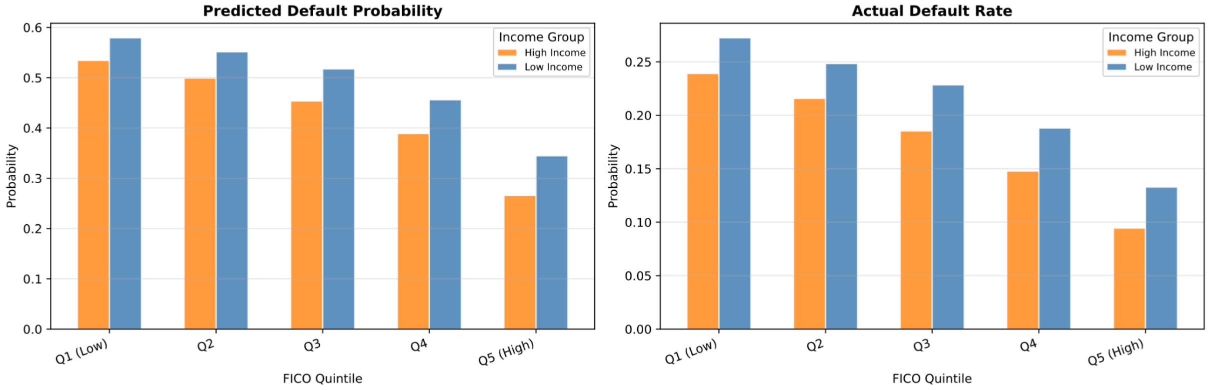

Figure 10: Conditional Fairness: Same FICO Bin, Different Income Group

Two patterns emerge from the comparison. First, the XGBoost model assigns higher predicted default probabilities to Low Income borrowers in every single quintile, even among borrowers with identical credit score ranges. The predicted gap widens steadily as creditworthiness increases, from 4.49 percentage points in Q1 to 7.90 percentage points in Q5: the largest income-based penalty falls precisely on the most creditworthy low-income borrowers.

Second, actual default rates also differ by income within each quintile, confirming that income carries genuine predictive information. In Q1, Low Income borrowers default at 27.2% versus 23.9% for High Income borrowers; in Q5, the gap narrows to 3.84 percentage points (13.3% versus 9.4%). Some degree of differential prediction is therefore empirically grounded. Table 5 reports the full breakdown across all quintiles.

FICO Quintile	Predicted Low	Predicted High	Actual Low	Actual High
Q1 (Low)	0.579	0.534	0.272	0.239
Q2	0.552	0.499	0.248	0.216
Q3	0.517	0.454	0.228	0.185
Q4	0.456	0.389	0.188	0.148
Q5 (High)	0.345	0.266	0.133	0.094

Table 5: Conditional Fairness Results

The excess bias metric disentangles what the model is justified in predicting from what it actually predicts. As defined in Section 4.3, excess bias is the difference between the predicted income gap and the actual income gap within each quintile. Figure 11 plots this quantity across the FICO distribution, and Table 6 reports the exact values.

FICO Quintile	Predicted Gap	Actual Gap	Excess Bias
Q1 (Low)	0.0449	0.0334	0.0116
Q2	0.0524	0.0326	0.0198
Q3	0.0637	0.0431	0.0206
Q4	0.0673	0.0402	0.0271
Q5 (High)	0.0790	0.0384	0.0406

Table 6: Excess Bias Gaps by FICO Quintile

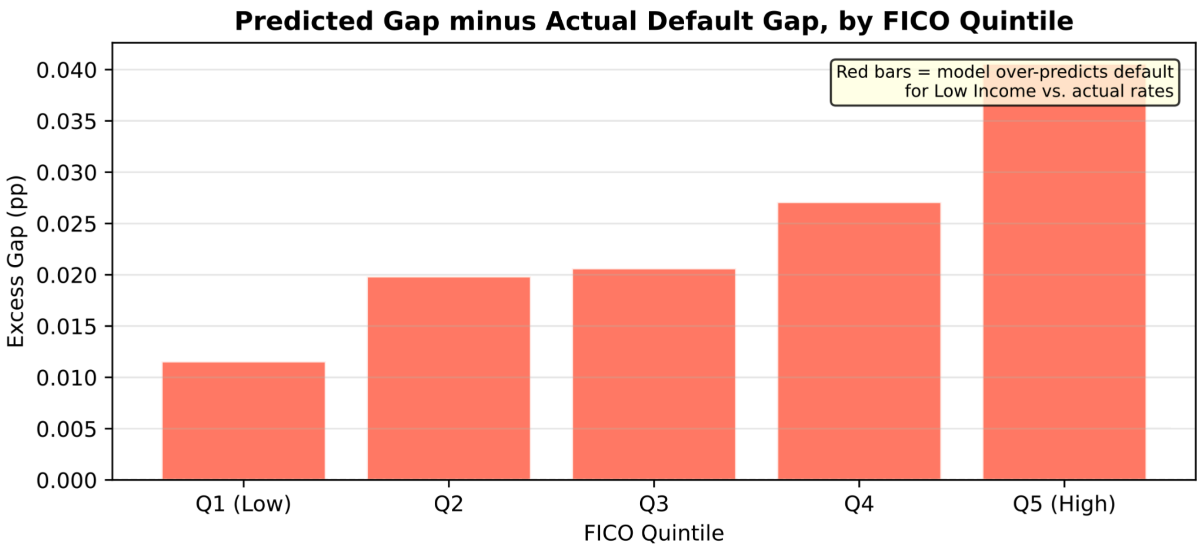

Figure 11: Excess Predicted Gap (Bias)

Excess bias is positive in every quintile and increases monotonically, from 1.16 percentage points in Q1 to 4.06 percentage points in Q5. This gradient is the most important result of the fairness audit: a model driven purely by credit risk should produce its smallest income-based penalty where the actual default gap is narrowest, yet the opposite occurs. As FICO scores rise and the observed default gap narrows, the model's unjustified penalty against low-income borrowers grows larger, suggesting that the algorithm compensates for the reduced discriminating power of FICO by leaning more heavily on income beyond what the data support. This is the pattern Chen (2024) identifies as algorithmic proxy discrimination: an output-level bias that no input audit would detect, because income is a legally permissible variable producing systematically discriminatory outcomes.

6. Limitations

6.1 Threats to Causal Validity

Four distinct threats to the validity of the Fuzzy RDD estimates are discussed here.

The most fundamental is the locality of the LATE. The IV2SLS estimator recovers the causal impact of interest rates on default only for borrowers whose assigned rate is materially affected by crossing FICO 740, roughly those in the range [715, 765]. This estimate cannot be assumed to generalise to borrowers at other points of the FICO distribution, and in particular not to borrowers near the entry threshold of FICO 660, who represent a qualitatively different population in terms of credit risk and financial fragility.

The weakness of the instrument ($F = 3.31$) means that the threshold indicator provides only limited exogenous variation in interest rates, producing IV2SLS estimates that are biased toward OLS with inflated standard errors. The non-significance of the LATE ($p = 0.751$) therefore does not constitute a clean null result; it reflects, at least in part, insufficient first-stage power rather than a true absence of a causal effect.

A central assumption of the RDD is that borrowers just above and just below the threshold are comparable on all dimensions other than the assigned interest rate. Figure 12 plots average log-transformed income against FICO score within the local bandwidth. The slope of the income-FICO relationship reverses at the cutoff: income rises with FICO score below 740 but falls above it, suggesting that the two groups differ not only in mean income but also in its relationship with creditworthiness. The formal t-test confirms a statistically significant income difference ($t = -2.452$, $p = 0.014$), which constitutes a direct violation of local comparability. Any income-related confounding, such as differential access to liquidity buffers, would bias the LATE upward.

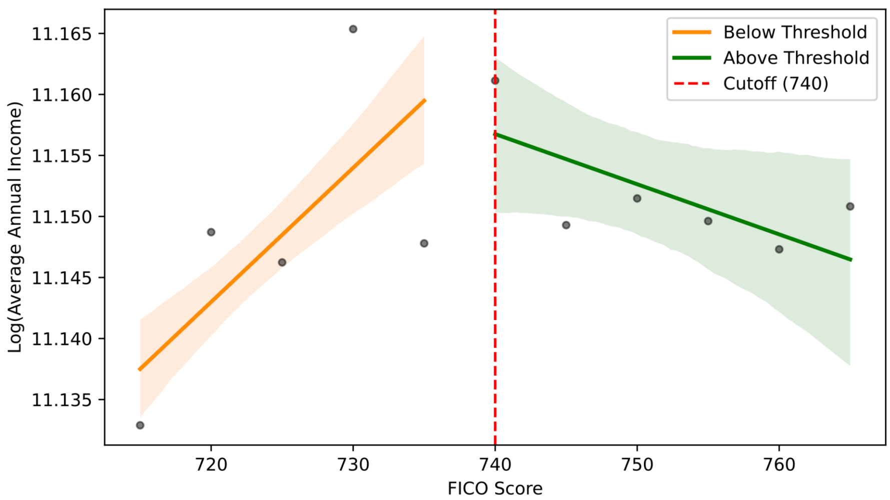

Figure 12: Covariate Balance : Log(Income) at Cutoff

Finally, the placebo analysis reveals that competing discontinuities in the LendingClub rate schedule complicate the isolation of FICO 740 as the unique source of exogenous variation. Significant rate jumps at FICO 720 and 760 are attributable to the platform's rate-setting grid, and the entry cutoff at FICO 660 generates a discontinuity of +7.73 percentage points, an order of magnitude larger than the first-stage estimate at 740. Future work employing a multi-cutoff RDD design could better account for this structure.

6.2 Methodological and Data Constraints

The income variable is drawn from borrower self-declarations and is not independently verified. If low-income borrowers over-report earnings more than high-income borrowers, some will be misclassified into the high-income group, pushing the estimated fairness gaps toward zero. The gaps documented in Sections 5.2 and 5.4 should therefore be read as conservative lower bounds.

The three-feature specification deliberately isolates the variables most relevant to the research questions, but understates predictive performance relative to real-world FinTech systems that use many more signals. As Fuster et al. (2022) show, the relationship between aggregate accuracy and group-level fairness becomes increasingly unpredictable as feature dimensionality grows, so it is unclear whether the documented gaps would increase or decrease in a richer model. As noted in Section 2.2, Barocas, Hardt, and Narayanan (2023) show that Demographic Parity, Equal Opportunity, and Predictive Parity cannot all be satisfied simultaneously when base rates differ across groups, and the choice among them is inherently normative. Finally, the dataset covers a single platform over 2007 to 2018, and the findings may not generalise to other FinTech lenders, credit products, or regulatory environments.

7. Discussion and Conclusions

7.1 Synthesis of Findings

This paper addressed two questions about FinTech lending: whether higher interest rates causally increase the probability of default, and whether machine learning credit models produce systematically biased predictions against low-income borrowers.

On the causal side, the results are suggestive but not conclusive. The FICO 740 threshold generates a visible shift in both interest rates and default probabilities, and the direction of the effect is consistent with what the debt-burden channel predicts: borrowers who pay more in interest have less money available to repay their loans. However, the instrument is too weak to support a firm causal claim, and a covariate imbalance at the cutoff introduces additional uncertainty. Pricing decisions are not merely reflections of risk; they may themselves be a source of it, but this paper cannot establish that with sufficient precision.

On the fairness side, the evidence is stronger and more troubling. All three classifiers show large fairness gaps detrimental to low-income borrowers, and the small improvement in predictive accuracy from logistic regression to XGBoost does not reduce these disparities at all. The SHAP analysis shows that income consistently pushes predicted default probabilities upward for low-income borrowers across the entire test set. Most importantly, the conditional fairness analysis shows that this income-based penalty is not justified by actual default behaviour: after holding FICO score fixed within quintiles, the model still over-predicts default risk for low-income borrowers, with an excess bias that grows from 1.16 to 4.06 percentage points as creditworthiness increases. The model penalises low-income borrowers most heavily precisely where that penalty is least justified.

Taken together, these findings point to a self-reinforcing feedback loop. A borrower who is over-predicted as high-risk receives a higher interest rate; that higher rate increases the probability of default; and that default reinforces the training signal that justified the over-prediction in the first place. The algorithmic system does not merely reflect pre-existing inequality in the data; it amplifies it in ways that are invisible to current regulatory frameworks.

7.2 Policy Implications

Current anti-discrimination law in consumer finance was designed to prevent lenders from explicitly using protected characteristics such as race or gender in credit decisions. This approach is not sufficient to address the problem documented here. The excess bias identified in Section 5.4 does not arise from the explicit use of a protected attribute; it arises because the model learns to use income, a legally permissible variable, as a proxy for socioeconomic group membership, producing discriminatory outcomes from seemingly neutral inputs.

The appropriate regulatory response is a shift from auditing what variables a model uses to auditing whether its predictions are conditionally biased after controlling for credit risk. The excess bias metric used in this paper offers a concrete and implementable template for this kind of output-side audit. Regulators could require lenders to compute and disclose this statistic as part of routine model validation, and to remediate models where excess bias exceeds a defined threshold. This would not prohibit the use of income as a predictive feature; it would simply require that any differential prediction be proportional to observed differences in actual default rates, a standard that the models evaluated here consistently fail to meet. Detecting algorithmic proxy discrimination is not enough: regulatory frameworks must be redesigned to require it routinely, as a condition of model deployment in consumer credit markets.

7.3 Directions for Future Research

Several extensions would strengthen both analyses. On the causal side, a multi-cutoff RDD design exploiting the full LendingClub rate grid as a system of instruments would recover a stronger first stage and allow more reliable causal estimates across the credit quality distribution. On the fairness side, training fairness-constrained variants of XGBoost and measuring the accuracy-bias trade-off would provide actionable guidance for platform operators. More broadly, a longitudinal design linking loan outcomes to model updates over time would allow direct empirical testing of the feedback loop described in Section 7.1, which this cross-sectional analysis can identify but not fully resolve.

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